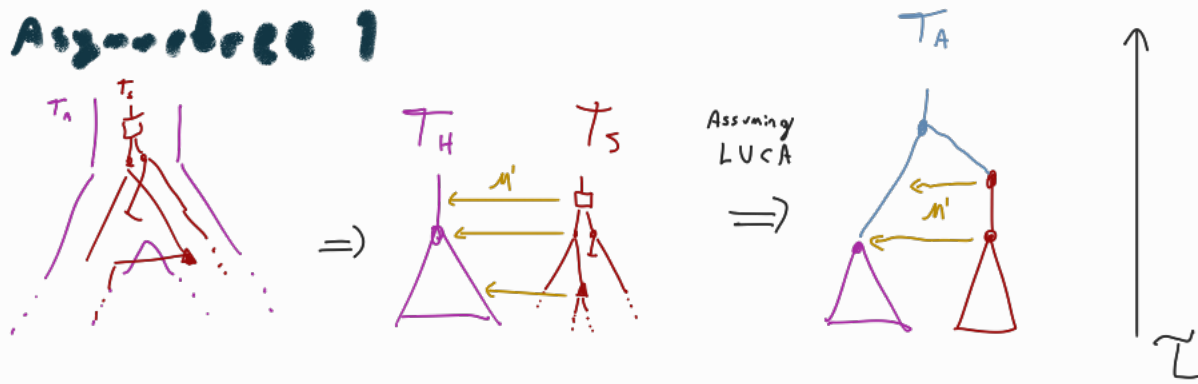


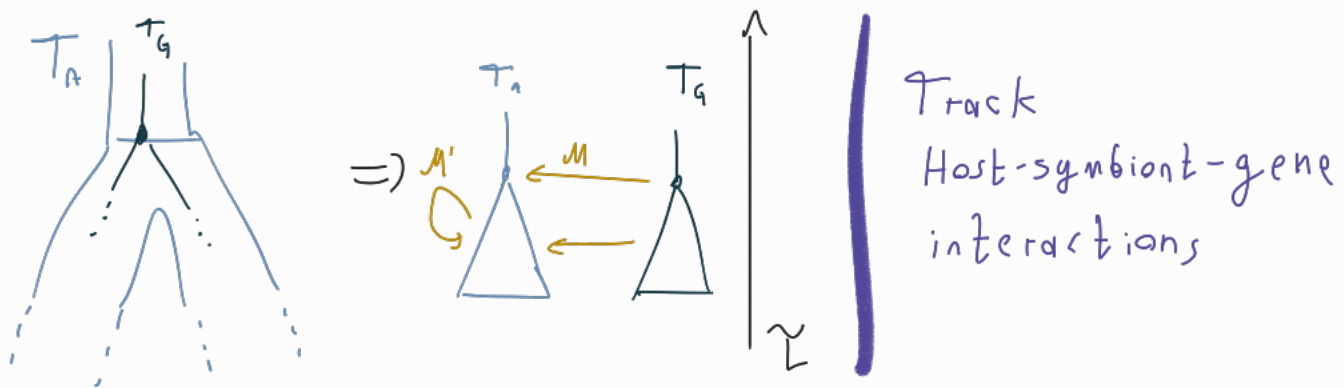
① simulate host-symbiont scenario

Algorithm 1



② simulate holobiont scenario

Algorithm 1



③ Return "complete" scenarios

Holobiont scenario

T_A, T_G, M, M', γ

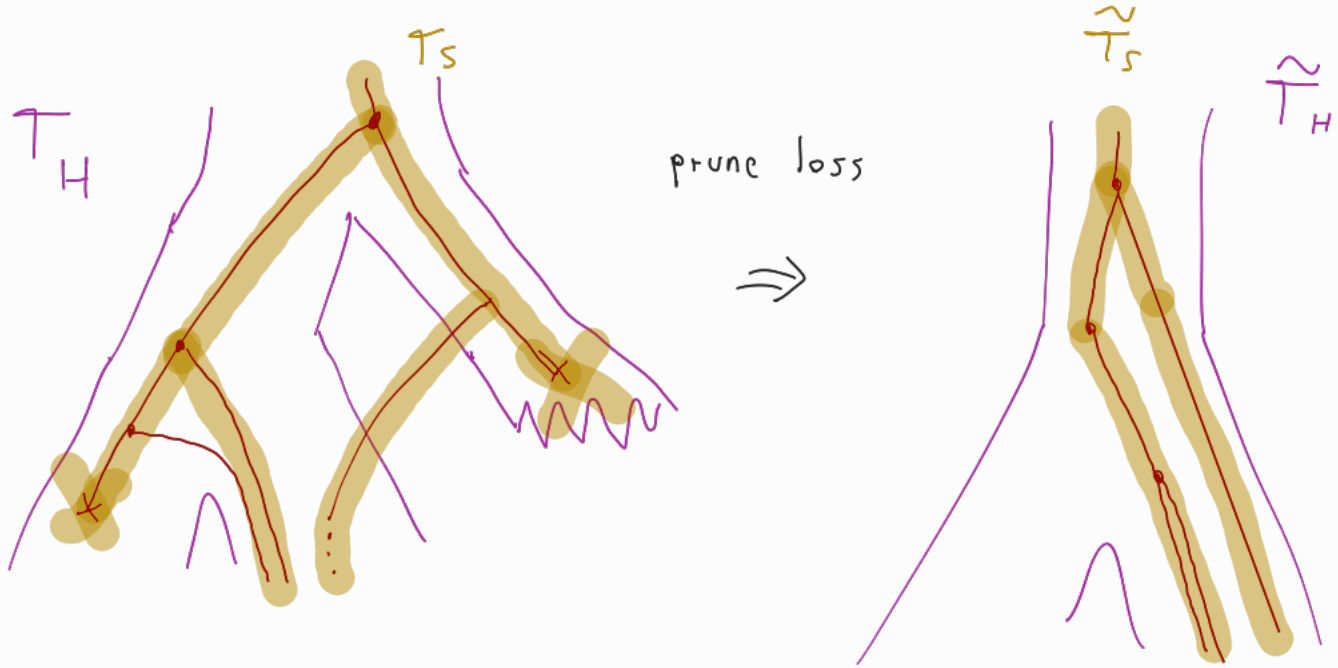
Two-level scenarios

T_H, T_S, M', γ

$T_G, T_S, M|_S, \gamma$

$T_G, T_H, M|_H, \gamma$

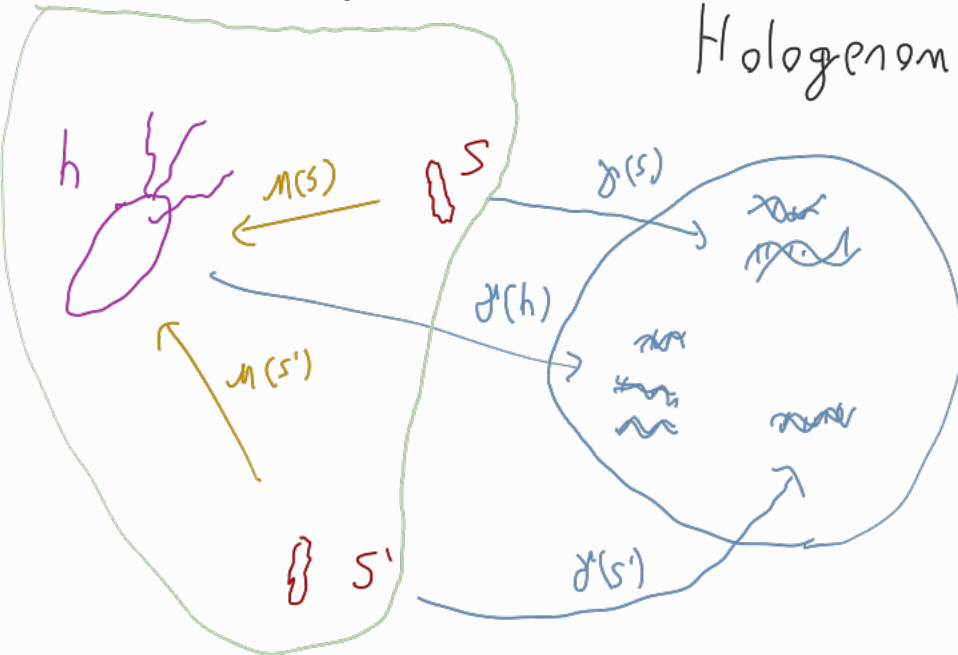
(4) Return "observable" scenarios
 pruned versions of restricted maps M, M'



(5) Gene-holobiont interaction

Holobiont system

Hologenome



Metabolic redundancy
increases loss rates

$$r_i^* = r_i + \alpha \frac{\text{\# Genes in host/symbiont}}{\text{Av. \# Genes in holobiont}}$$

User-provided
Normalization factor

(6)

Only-inter transfer



Gene transfer between species a, b is allowed only if

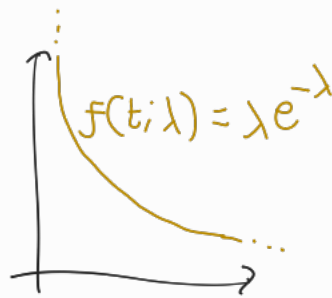
$$\mu'(a) = \mu'(b)$$

(7)

Current version of Asymmetree

Waiting time for evolutionary event is drawn from exponential distribution

Asymmetree 1



$$\lambda = \underbrace{t_d + t_i + t_b + t_s + t_c}_{\text{Event rates}}$$